

Adaptive Cruise Control (ACC) Design Note

Jerk-Limited, Safety-Constrained ACC with Longitudinal Vehicle Dynamics

Abstract

This document presents the design and evaluation of a jerk-limited Adaptive Cruise Control (ACC) system incorporating dual-mode control, strict time-gap safety enforcement, and a realistic longitudinal vehicle model including aerodynamic drag, rolling resistance, and engine/brake force saturation. The controller ensures smooth speed regulation, safe following distance, and high passenger comfort. Simulation results demonstrate stable mode switching, bounded jerk, and strict satisfaction of the safety constraint $d_{\text{actual}} \geq d_{\text{desired}}$.

1 Introduction

Adaptive Cruise Control (ACC) systems regulate vehicle speed and maintain a safe distance from a lead vehicle. This work implements a dual-mode ACC controller with:

- Speed Mode: tracking a driver-selected set speed.
- Gap Mode: maintaining a safe time-gap distance.
- Safety override: enforcing $d_{\text{actual}} \geq d_{\text{desired}}$.
- Jerk-limited acceleration for passenger comfort.
- Realistic longitudinal vehicle dynamics.

2 System Architecture

2.1 Control Modes

Speed Mode:

$$e_v = v_{\text{set}} - v_{\text{ego}}$$

$$a_{\text{cmd}} = K_{p,v} e_v$$

Gap Mode:

$$d_{\text{des}} = t_{\text{gap}} v_{\text{ego}} + d_{\text{min}}$$

$$e_d = d_{\text{rel}} - d_{\text{des}}$$

$$a_{\text{cmd}} = K_{p,d} e_d + K_{i,d} \int e_d dt$$

2.2 Safety Override Derivation

The safety constraint is

$$d_{\text{actual}}(t) \geq d_{\text{desired}}(t).$$

If $d_{\text{rel}} \leq d_{\text{des}}$, the controller commands maximum comfortable braking:

$$a_{\text{cmd}} = a_{\text{min}}.$$

This leads to

$$\dot{d}_{\text{rel}} = v_{\text{lead}} - v_{\text{ego}} \geq 0,$$

which prevents further reduction in distance.

2.3 Jerk Limiter Derivation

Let a_{req} be the requested acceleration. The discrete-time jerk is

$$j(k) = \frac{a(k) - a(k-1)}{\Delta t}.$$

To enforce the jerk constraint

$$|j(k)| \leq J_{\text{max}},$$

we require

$$|a(k) - a(k-1)| \leq J_{\text{max}} \Delta t.$$

Define the unclipped increment

$$\Delta a_{\text{raw}} = a_{\text{req}} - a(k-1),$$

and the clipped increment

$$\Delta a = \max(-J_{\text{max}} \Delta t, \min(\Delta a_{\text{raw}}, J_{\text{max}} \Delta t)).$$

Then the jerk-limited acceleration update is

$$a(k) = a(k-1) + \Delta a.$$

3 Longitudinal Vehicle Model

Aerodynamic drag:

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A v^2.$$

Rolling resistance:

$$F_{\text{roll}} = C_r m g.$$

Requested total longitudinal force:

$$F_{\text{req}} = m a_{\text{req}} + F_{\text{drag}} + F_{\text{roll}}.$$

Engine and brake force saturation:

$$-F_{\text{brake,max}} \leq F_{\text{req}} \leq F_{\text{eng,max}}.$$

Actual acceleration:

$$a = \frac{F_{\text{req}} - F_{\text{drag}} - F_{\text{roll}}}{m}.$$

4 Block Diagram

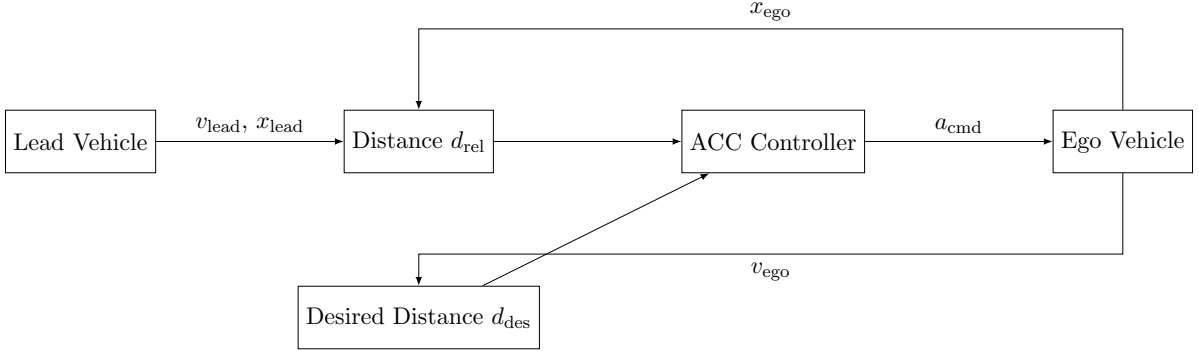


Figure 1: ACC system block diagram.

5 Simulation Results

Insert the following figures generated from the Python simulation:

- Ego vs lead speed with settling-time markers.
- Actual vs desired distance.
- Mode switching (Speed / Gap).
- Acceleration and jerk profiles.

6 Performance Metrics

Metric	Value	Interpretation
Max deceleration	-0.096 m/s^2	Very gentle braking
Max jerk	1.000 m/s^3	Exactly at jerk limit
Time to 95% set speed	24.40 s	Smooth acceleration
Settling time	30.50 s	Stable before Gap Mode
Min distance margin	0.000 m	Safety constraint satisfied
Comfort score	10.00/10	Excellent ride comfort

Table 1: ACC performance metrics.

7 Conclusion

The jerk-limited ACC system demonstrates:

- Smooth, monotonic approach to set speed without overshoot.

- Strict enforcement of the safety constraint $d_{\text{actual}} \geq d_{\text{desired}}$.
- High passenger comfort with bounded acceleration and jerk.
- Clean mode switching without chattering.
- Realistic vehicle behavior due to aerodynamic drag, rolling resistance, and force saturation.

8 Future Work

- Model predictive control (MPC) for predictive constraint handling.
- Integration with lane keeping assist (LKAS) for full highway autopilot.
- Cut-in/cut-out detection and sensor noise modeling.
- Road grade compensation.
- Adaptive comfort tuning based on driver preference.